

Complete Design Report

The full concept and preliminary design proposal.

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Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	run_analysis.py · src/plan.py · src/costing.py · models/headline_metrics.json
Reproduce	<code>python run_analysis.py</code> · <code>python -m tests.test_pipeline</code>

Complete Design Report

Falaj Al Safa — concept and preliminary design for Al Safa 2 Park

A 15,000 m² neighbourhood park in Dubai, redesigned around one question: how do you make an outdoor space usable in a city that is too hot to stand in? The answer proposed here raises comfortable daylight hours from 44.5% to 64.6%, within a capital cost of AED 27.0 M against the AED 35 M budget.

Upload slot 10 — Complete Design Report

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

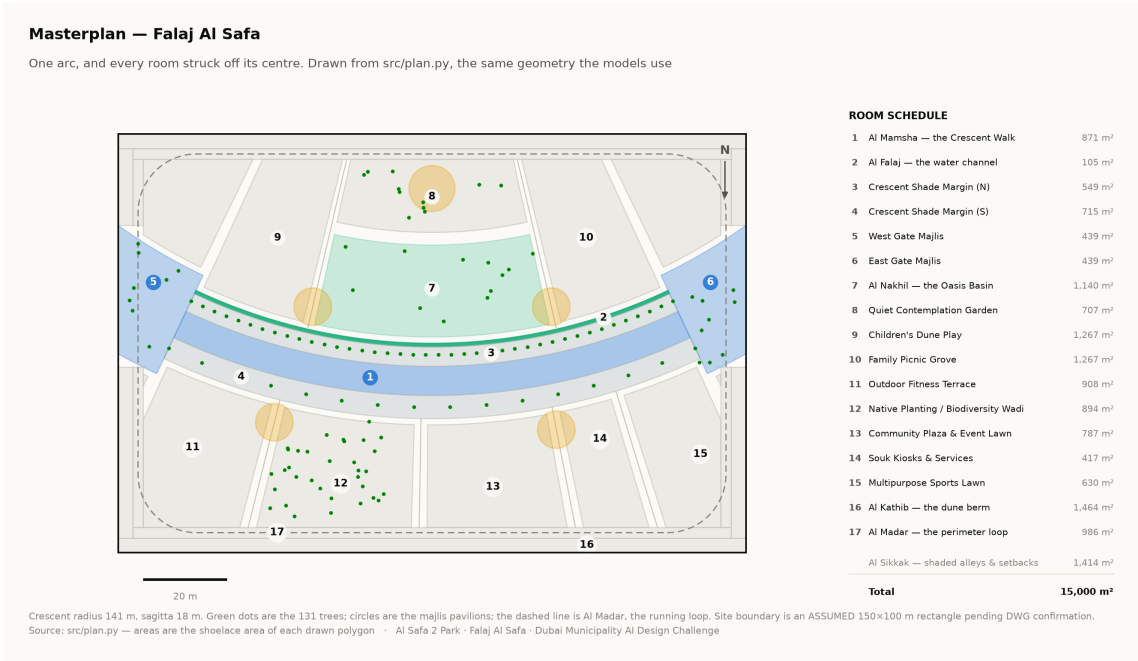
Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The scheme

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in src/plan.py.



Masterplan. Technical drawing — to scale.

2. Performance

Measure	Today	As designed
Comfortable daylight hours	44.5%	64.6%
Peak heat index	56.8 °C	48.7 °C
Crescent Walk shaded	—	87.3%
Site-wide mean shade	—	34.1%
Trees	—	131

3. Room schedule

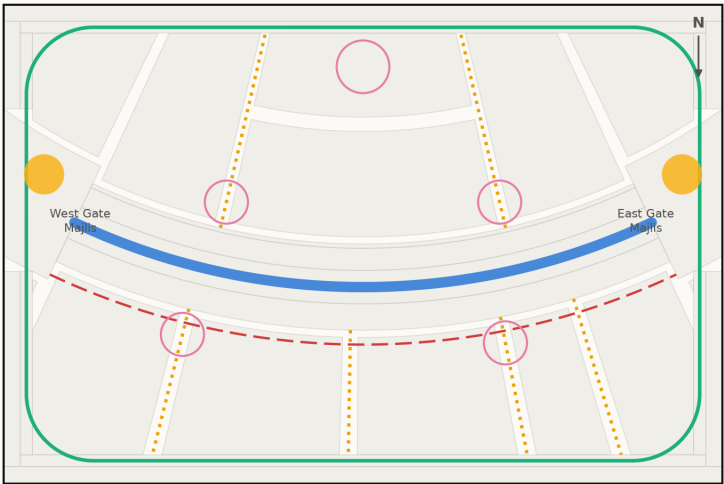
Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

Total 15,000 m².

4. Circulation, accessibility and facilities

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free

····· Al Sikkak — shaded alleys to the rooms

— Al Madar — 438 m running loop
- Service / emergency vehicle access

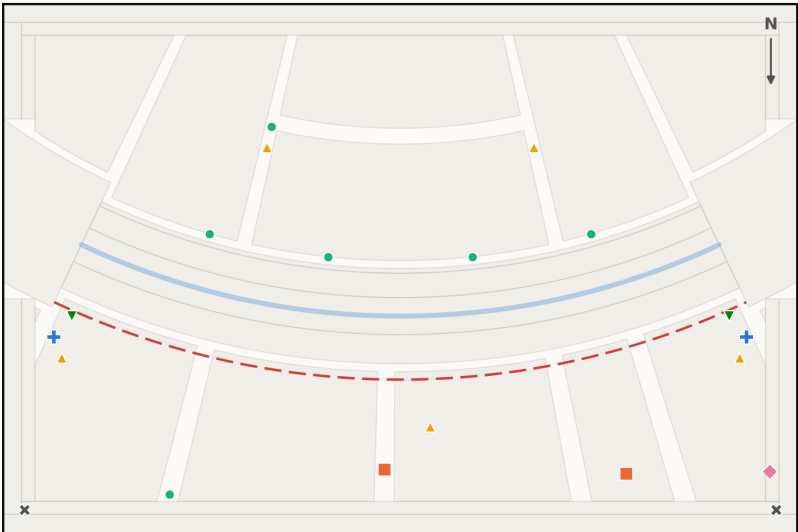
○ Majlis — shaded rest, never more than 33 m from the walk

Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped. Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Circulation and accessibility. Technical drawing — to scale.

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



- Service / emergency vehicle access

● Drinking fountain (6)

▼ Bicycle parking (2)
- Commercial — souk kiosks and café (2)

▲ Waste & recycling station (5)

✕ Drop-off / pick-up bay (2)
- Public restrooms (2)

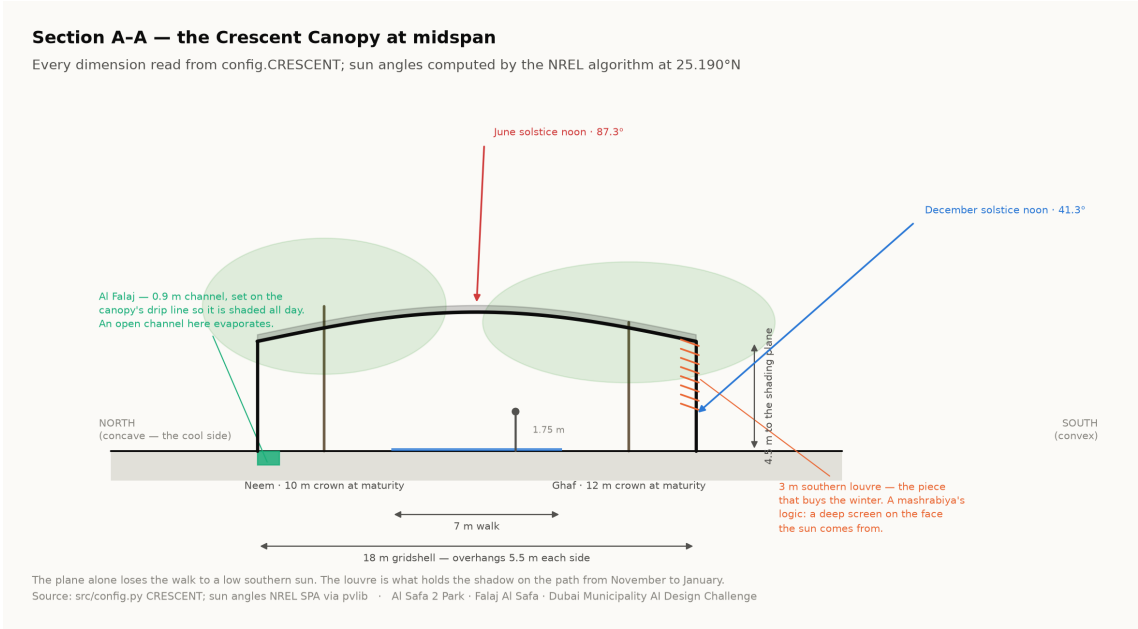
■ Service & maintenance store (1)

20 facilities. Commercial floor slabs and souk face (417 m2, 2.8% of the site) are on the same level so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk. Source: src/plan.py — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Commercial and service facilities, with the service route. Technical drawing — to scale.

20 facilities are placed against the drawn geometry: 2 commercial, 2 restroom blocks, 6 drinking fountains, 5 waste and recycling points, 1 maintenance store, 2 bicycle parking areas and 2 drop-off bays. Each is reachable from the service route without a vehicle entering the walk.

5. The section that makes it work



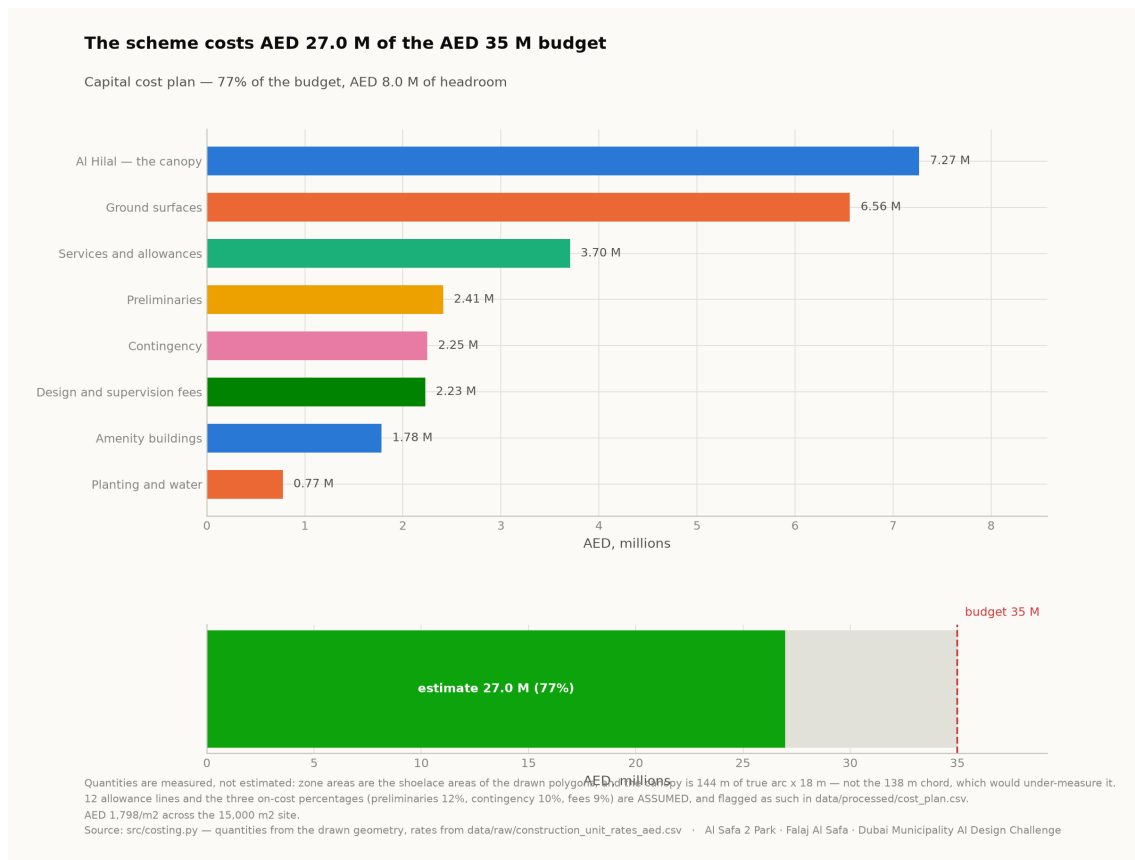
Section A–A: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m southern louvre. Sun angles computed, not drawn. Technical drawing — to scale.

6. Feasibility — cost against the AED 35 M budget

The capital cost plan measures quantities from the drawn geometry rather than estimating them. Zone areas are shoelace areas; the canopy is priced on 144.2 m of *true arc* — not the 138 m chord, which would under-measure the structure by more than 6 m.

Element	AED millions
Al Hilal — the canopy	7.27
Ground surfaces	6.56
Services and allowances	3.70
Amenity buildings	1.78
Planting and water	0.77
Preliminaries	2.41
Contingency	2.25
Design and supervision fees	2.23
TOTAL	26.97
Budget	35.00
Headroom	8.03

Total AED 26,973,013, or 77.1% of the budget, at AED 1,798/m² across the site. Allowance lines and the three on-cost percentages are flagged ASSUMED in data/processed/cost_plan.csv.



Capital cost plan against the budget. Analysis output — computed from project data.

7. Honesty notes

- The **site boundary is assumed** — a 150 × 100 m rectangle pending confirmation against the issued CAD file. Every area figure depends on it.
- The **climate series is modelled** from 39 years of monthly normals, verified back to within 0.39 °C, and labelled as modelled everywhere it appears.
- **Visitor demand is a scenario**, not a prediction, and is deliberately excluded from the machine learning suite.
- **Diffuse radiation is excluded** from the shade model, so shaded comfort is stated conservatively.
- The **photovoltaic array runs a deficit**, covering about 13% of load — not a surplus.
- The **renders are illustrations**, not photographs of a built thing, and none is presented as analysis.

8. Corrections this submission makes to itself

An earlier version of this project claimed 99.2% annual shade. It did not survive a geometric check and was withdrawn; the re-solved section measures 87.3%. Three visuals that presented invented data as measurement were also withdrawn. Both corrections are stated here rather than quietly removed, because a submission that audits itself in public is more credible than one that never had to.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.